

# **BLAST\_FURNACE\_SHUTDOWN\_SOP**

STEEL AUTHORITY OF INDIA LIMITED

BOKARO STEEL PLANT

Standard Operating Procedure

SOP No: BSP-BF-001

Title: Planned Shutdown Procedure for Blast Furnace

Department: Iron Making

Revision: 02

Date: 15-Jan-2024

## **1. PURPOSE**

This SOP defines the step-by-step procedure for executing a planned shutdown of the Blast Furnace at Bokaro Steel Plant. It ensures the safety of personnel, protection of equipment, and systematic wind-down of operations.

## **2. SCOPE**

Applicable to Blast Furnaces No. 1, 2, 3, and 4 at Bokaro Steel Plant.

This SOP must be followed by all shift managers, furnace operators, and maintenance personnel involved in the shutdown process.

## **3. PREREQUISITES**

- Written shutdown authorization from General Manager (Iron Making)
- All maintenance teams on standby
- Emergency response team notified
- All hot metal ladles cleared from the cast house floor
- Gas cleaning plant in operational condition

## **4. PERSONAL PROTECTIVE EQUIPMENT (PPE) REQUIRED**

- Heat-resistant suit (minimum 1000°C rating)
- Face shield with IR filter (shade 5 or higher)
- Leather gauntlet gloves
- Safety boots with steel toe cap
- Self-contained breathing apparatus (SCBA) if entering tuyere area

## **5. SHUTDOWN PROCEDURE**

Step 1 — Reduce Coke Rate

- Gradually reduce coke rate by 10% every 30 minutes
- Monitor furnace top pressure continuously
- Target: reduce blast volume to 60% of normal operating level
- Duration: approximately 2 hours

Step 2 — Stop Ore Charging

- Stop iron ore and sinter charging from stock house
- Continue coke charging to maintain thermal balance

- Notify stock house supervisor of charging stoppage
- Record last charge number in the furnace log

#### Step 3 — Reduce Hot Blast Temperature

- Reduce stove temperature from operating level (1150°C) to 900°C
- Rate of reduction: not more than 50°C per 30 minutes
- Monitor thermal state of furnace continuously
- If furnace top temperature exceeds 350°C, halt reduction temporarily

#### Step 4 — Final Cast

- Conduct final hot metal cast to empty the hearth
- Ensure all hot metal is transferred to torpedo ladles
- Record final hot metal temperature (target: minimum 1480°C)
- Complete slag removal from slag notch

#### Step 5 — Cut Off Hot Blast

- Close all tuyere valves sequentially (start from tuyere No. 1)
- Close bustle pipe isolation valve
- Switch off all blowers
- Notify gas cleaning plant to prepare for low-load operation

#### Step 6 — Isolate Gas System

- Close furnace top gas offtake valve
- Purge gas lines with nitrogen (minimum 3 purge cycles)
- Verify CO level below 20 ppm using portable gas detector before entry
- Lock out and tag out (LOTO) all gas isolation valves

#### Step 7 — Cool Down

- Allow natural cooling for minimum 48 hours before any internal inspection
- Monitor shell temperature every 4 hours
- Do not allow water quenching unless approved by Chief of Operations

### 6. POST-SHUTDOWN CHECKS

- Verify all gas valves are locked out
- Confirm CO level below 20 ppm at all entry points
- Check cooling water flow to all staves is maintained
- Submit shutdown completion report to shift manager within 2 hours

### 7. EMERGENCY CONTACTS

- Shift Manager (Iron Making): Extension 2201
- Gas Safety Officer: Extension 2310
- Plant Emergency Control Room: Extension 100

### 8. REFERENCES

- BSP Safety Manual, Chapter 7: Blast Furnace Operations
- IS 15656: Safety in Iron and Steel Industry